



Compensation of hysteresis in piezoelectric actuators without dynamics modeling

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ARTICLE INFO

Article history:

Received 6 December 2012

Received in revised form 26 April 2013

Accepted 26 April 2013

Available online 21 May 2013

Keywords:

Piezoelectric actuator

Hysteresis

Modeling

Nonlinear control

ABSTRACT

Along with the fast development of ultra-precision motion systems, model and control of piezoelectric actuators draw significant research interest. Unfortunately, one of the main obstacles hindering their applications is the hysteresis, which is not ignorable in applications that require a high accuracy in motion control. In this paper, a new NARMAX model based on BP neural network is proposed for modeling the nonlinear hysteresis behavior in piezoelectric actuator. Because the proposed model is constructed in an online way, there is no need to conduct experiments for parameters identification. In order to demonstrate the precision and the rate-dependent property of the proposed model, experiments are performed under designed excitations with different amplitudes and frequencies. Furthermore, taking advantage of the proposed model, we design a nonlinear controller based on adaptive inverse control for the compensation of hysteresis in the piezoelectric actuator. A feature of the developed controller is that it allows piezoelectric actuator which exhibits complex nonlinear hysteresis behavior to be directly controlled without dynamics modeling. Thus the developed controller can be generally used for various piezoelectric actuators with different dynamics, leading to favorable convenience in industrial applications. Because of essentially open-loop characteristics of the proposed control system, the instable phenomenon aroused by feedback can be avoided. Experiments are conducted to validate the performance of the developed control system. It is shown that the developed controller not only compensates the hysteresis in the piezoelectric actuator effectively, but also exhibits rate-dependent property that allows piezoelectric actuator to track both periodic and non-periodic motions accurately.

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1. Introduction

Piezoelectric actuators are widely employed in ultra-precision motion systems because of their distinctive advantages over other competing techniques, i.e. infinite displacement resolution, high stiffness, high actuating force and little heat generation. These versatile positioning elements have extensive applications in both industrial and academic research, including nanopositioning stage, optical fiber alignment, mask alignment, ultra-precision machining, active vibration control, scanning tunneling microscopy, hard disk drive and diamond turning machines [1–3].

However, since the materials of piezoelectric actuator are generally ferroelectric, piezoelectric actuator usually exhibits nonlinear hysteresis behavior in response to an applied electric field. Because hysteresis is a set-value nonlinear phenomenon whose presence is a kind of non-local memory, it causes positioning errors which significantly hinder the operating speed and precision of

piezoelectric actuator [4]. Although there is an almost linear relationship between the output displacement and input charge of piezoelectric actuator, charge control has not been widely used due to the complex circuitry of the power driver and associated reduction of the displacement detecting sensitivity [5]. Therefore, the voltage-driven strategy based on hysteresis compensation is prevalent in the control of piezoelectric actuator. The improvement of the linearity of piezoelectric actuator can be realized by implementing model-based compensation strategies. Several mathematical models of hysteresis have been proposed. Specifically, Ge and Jouaneh [6] modeled hysteresis nonlinearity using a modified generalized Preisach model and used this approach in a linearizing control scheme. However, this model required large quantity of databases and bulky calculation. Moreover, the Preisach model is a kind of rate-independent model which was not able to describe the rate-dependent hysteresis. Mokaberi and Requicha [7] formulated hysteresis using superposition of fundamental operators, and their inverse model was obtained by using the inversion properties of the Prandtl–Ishlinskii operators. Another well-known model in describing hysteresis is the Bouc–Wen model, originally proposed by Bouc [8] and later generalized by Wen [9]. Since only

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one auxiliary nonlinear differential equation is required to describe hysteresis behavior, the Bouc–Wen model features computational simplicity. Despite the straightforward and analytical properties, the symmetric structure and rate-independent property of the Bouc–Wen model limit its modeling accuracy. Since the reconstruction of the weight functions and the repeated identification of the model parameters needs plenty of experiments data and bulky calculation, modeling methods mentioned above lack ability of online update for adapting the variation of the plant. Thus, it is necessary to establish a general online hysteresis model which can be used for various piezoelectric actuators with different dynamics.

Hysteresis compensation technique has drawn extensive research interest because it plays a critical role in further improving the application of piezoelectric actuator in ultra-precision motion system. Gu et al. [10] presented a real-time inverse hysteresis compensation method for piezoelectric actuators exhibiting asymmetric hysteresis effect. The proposed method directly utilized a modified Prandtl–Ishlinskii hysteresis model to characterize the inverse hysteresis effect. Yi et al. [11] considered hysteresis nonlinearity of the actuator as a disturbance over a linear system. They presented a closed-loop compensation mechanism, which utilized a disturbance observer to estimate and compensate hysteresis nonlinearity. Jayawardhana et al. [12] provided a PID controller design for set point regulation and disturbance rejections in the context of second-order systems with hysteresis components. Rakotondrabe [13] proposed multiplicative inverse structure, based on which a proposed compensator scheme was adapted to hysteresis which was expressed by the Bouc–Wen model. Liu et al. [14] developed a model reference adaptive control scheme based on hyper stability theory for a moving stage system driven by a piezoelectric actuator. Gaul and Becker [15] proposed a model-based approach for the compensation of the piezoelectric hysteresis and creep effects in a feedforward control design. The proposed control design combined a flatness-based inversion procedure for structural dynamics under the assumption of linear piezoelectricity and an inverse piezoelectric operator for compensating hysteresis. Li and Xu [16] proposed an sliding mode control with perturbation estimation featuring a PID-type sliding surface for the motion tracking control of a micro-manipulator system with piezoelectric actuator. Leang and Devasia [17] reported a design of a high-gain feedback controller using a notch filter to improve the stability margin to linearize hysteresis. It is worth mentioning that in previous literatures, whether used feedforward control scheme or feedforward control combined with feedback control scheme, the design of control system depended on precise hysteresis model. If the piezoelectric actuator is substituted for another one which exhibits a different hysteresis behavior, a series of works are required. First of all, hysteresis behavior in the new piezoelectric actuator needs to be measured through experiments. Then, the parameters of these models should be identified. The genetic algorithm (GA) [18] or the particle swarm optimization (PSO) [19], whichever we resort to, the parameters identification is a laborious work. In the end, the redesign of the controller based on the newly constructed model needs to be implemented. It is obvious that these kinds of control frames lack a certain of flexibility and may result in low designing efficiency in industrial application.

In the present work, a nonlinear autoregressive moving average with exogenous inputs (NARMAX) model based on backpropagation (BP) neural network is proposed to describe hysteresis behavior in piezoelectric actuator. Because the proposed model is constructed in an online way, there is no need to conduct experiments for parameters identification. In order to demonstrate the precision and the rate-dependent property of the proposed model, experiments are performed under excitations with different amplitudes and frequencies. Furthermore, taking advantage of the proposed model, we design a nonlinear controller based on adaptive inverse control in order to compensate hysteresis

in piezoelectric actuator. A distinctive feature of the developed controller is that it allows piezoelectric actuator which exhibits complex nonlinear hysteresis behavior to be directly controlled without dynamics modeling. Experiments were conducted to validate the performance of the proposed hysteresis model and the developed control system.

2. Mathematical model of hysteresis

2.1. NARMAX model based on BP neural network

The NARMAX model is a black-box nonlinear model which provides a powerful representation for time series analysis, modeling and prediction due to its capability of accommodating the dynamic, complex and nonlinear nature of time series prediction problems [20]. Because of powerful capability of nonlinear modeling, the NARMAX model is suitable for hysteresis modeling and compensation of piezoelectric actuator. A special case of the general NARMAX model is represented as follows [21]:

$$y(t) = F[y(t-1), \dots, y(t-n_y), u(t-1), \dots, u(t-n_u)] + e(t), \quad (1)$$

where $y(t)$, $u(t)$ and $e(t)$ represent the output signal, the input signal and the system noise, respectively; n_y , n_u and n_e are the associate maximum lags and $F(\cdot)$ is the unknown nonlinear function. Assuming that $e(t)$ has zero mean and a finite variance, an optimal predictor for NARMAX model in (1) is approximately given by

$$\hat{y}(t) = F[y(t-1), \dots, y(t-n_y), u(t-1), \dots, u(t-n_u)]. \quad (2)$$

Consider both the accuracy and efficiency of calculation, we exploit the optimal predictor to describe hysteresis behavior in piezoelectric actuator. In this case, the unknown nonlinear function $F(\cdot)$ in (2) is modeled by the backpropagation (BP) neural network.

The starting point of BP neural network is imitating how human brain works. It has a layered structure and each layer consists of a series of adaptive linear neurons (ADALINES). The BP network is basically a gradient decent algorithm designed to minimize the error function in the weights space. During training of the neural network, weights are adjusted to decrease the total error [22].

It follows from (2) that the NARMAX model has self-feedback property. Thus, the input of neural network should contain its previous outputs. Actually, the real intelligence of the neural network exists in weights in ADALINES. For this neural network, we employ BP learning algorithm to adjust the weights in ADALINES. The BP learning algorithm takes iterative steps: first, input signal is applied to the network which produces output signal based on the current state of ADALINES' weights; then, this output signal is compared with the known output signal, and a mean-squared error signal is calculated; next, the error value is propagated backward through the neural network and changes are made to the weights in order to reduce the error signal in each layer. Although BP learning algorithm can be applied to the neural network with any number of layers, actual experiments showed us that one hidden layer can reach sufficient precision and owns preferable efficiency.

2.2. Online modeling of rate-dependent hysteresis

The dynamics of piezoelectric actuator can be explained by a general electromechanical model, which includes the electrical aspect and the mechanical aspect, as well as their mutual interaction [23,24]. It is known that hysteresis can be classified into rate-independent hysteresis and rate-dependent hysteresis. The rate-independent hysteresis, whose behavior depends only on the input extrema, is also known as static hysteresis. By comparison, the rate-dependent hysteresis, whose behavior related with the amplitude, frequency and phase of the applied voltage, is also

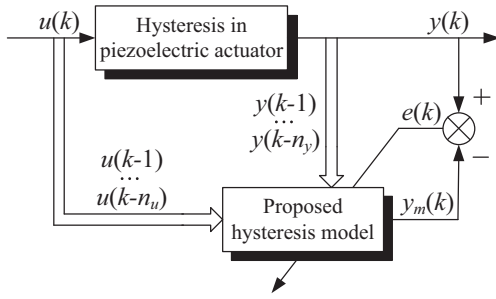


Fig. 1. The scheme of the modeling process for the proposed hysteresis model.

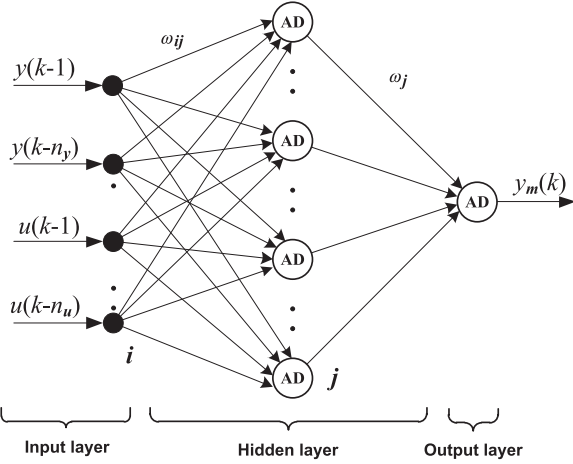


Fig. 2. The structure of online adaptive rate-dependent hysteresis model.

known as dynamic hysteresis. Obviously, it is more complex to model rate-dependent hysteresis than rate-independent hysteresis which is an intrinsic property of the piezoelectric actuators [25].

The main challenge in modeling rate-dependent hysteresis is to capture the variation of the input-to-output mapping. In this case, the input signal is the voltage applied to piezoelectric actuator, and the output signal is the displacement of piezoelectric actuator. Because hysteresis behavior in piezoelectric actuator depends not only on the instantaneous value of the applied voltage but also on the history of operational process, exploiting the NARMAX model which contain previous information to model the rate-dependent hysteresis is reasonable. Thus, we employed NARMAX model based on BP neural network to construct an online adaptive rate-dependent hysteresis model for the piezoelectric actuator. The scheme of the modeling process for the proposed hysteresis model is shown in Fig. 1, where k represents the k th sampling time; $u(k)$ and $y(k)$ represent the applied voltage and the measured displacement of piezoelectric actuator, respectively. We define $X(k) = \{y(k-1), \dots, y(k-n_y); u(k-1), \dots, u(k-n_u)\}$ as the input vector which contains the previous applied voltages and measured displacements of piezoelectric actuator. The output of the proposed hysteresis model $y_m(k)$ is compared with the measured displacement $y(k)$ of piezoelectric actuator, and then the error $e(k)$ is fed back to the proposed hysteresis model for update. The structure of the online adaptive rate-dependent hysteresis model and the adaptive linear neuron are shown in Figs. 2 and 3, respectively.

In this case, we define the limiting function in ADALINES of the hidden layer and output layer as log-sigmoid function, that is

$$f(x) = \frac{1}{1 + e^{-x}}. \quad (3)$$

The log-sigmoid function is used in neural network application because it is mathematically well behaved, differentiable and

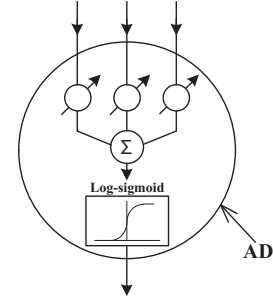


Fig. 3. The adaptive linear neuron.

is a strictly increasing function, which reduces the computational requirements during training [26].

As mentioned above, the input signal of the proposed hysteresis model broadcasts in the neural network from input layer to output layer. We define the outputs of input layer as:

$$O_i(k) = X_i(k), \quad i = 1, 2, \dots, n_y + n_u. \quad (4)$$

It follows from Fig. 2 and (3) that the output of the hidden layer can be derived as:

$$O_j(k) = f \left[\sum_{i=1}^{n_y+n_u} \omega_{ij}(k-1) O_i(k) \right], \quad j = 1, 2, \dots, Q \quad (5)$$

where Q and ω_{ij} represent the number and the weights of ADALINES in the hidden layer, respectively.

According to (3) and (5), the output of the output layer can be derived as:

$$y_m(k) = \sum_{j=1}^Q \omega_j(k-1) O_j(k) \quad (6)$$

where ω_j represents the weights in ADALINES in output layer. It follows from (6) that the error between the measured displacement of piezoelectric actuator and the proposed hysteresis model can be obtained as:

$$e(k) = y(k) - y_m(k) \quad (7)$$

At the k th sampling time, the weights in output layer can be computed as:

$$\omega_j(k) = \omega_j(k-1) + \eta e(k) O_j(k) + \alpha (\omega_j(k-1) - \omega_j(k-2)) \quad (8)$$

where η is defined as a positive number which represents the study rate of $\omega_j(k)$; α represents a dynamic term whose value varies on the interval $[0, 1)$. Adding this dynamic term can accelerate the convergence by considering previous experience. Using the updated weights $\omega_j(k)$, we compute the weight in the hidden layer at the k th sampling time as:

$$\begin{aligned} \omega_{ij}(k) = & \omega_{ij}(k-1) + \eta e(k) \omega_j(k-1) \cdot f' \left[\sum_{i=1}^{n_y+n_u} \omega_{ij}(k-1) O_i(k) \right] \\ & \times O_i(k) - \alpha [\omega_{ij}(k-1) - \omega_{ij}(k-2)] \end{aligned} \quad (9)$$

In order to allow the proposed online adaptive rate-dependent hysteresis model own high precision as well as satisfactory calculation efficiency, we need to determinate its suitable structure. After a process of trial and error, we finally specified the maximum lags of output signal n_y and the maximum lags of input signal n_u as 1 and 0, respectively; in addition, the number of ADALINES in the hidden layer Q is chosen as 20.

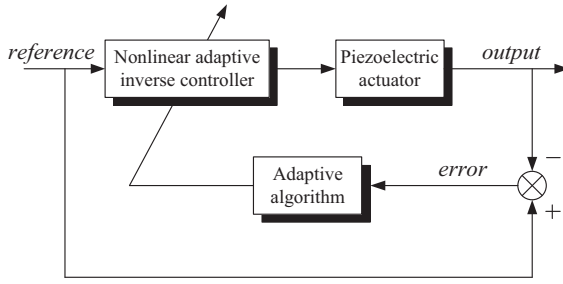


Fig. 4. The structure of adaptive inverse control for piezoelectric actuator.

3. Adaptive inverse control of piezoelectric actuator

3.1. Structure of control system

Controllers for piezoelectric actuator were traditionally designed based on hysteresis model which had already been constructed. During these kinds of modeling process, a series of experiments needed to be conducted for parameters identification. It seems that these major steps become indispensable in the standard flow process of controller design for piezoelectric actuator.

This paper considers adaptive inverse control, which is based on the concept of dynamic inversion, but an inverse needs not exist [27]. In this case, control of piezoelectric actuator is achieved by preceding it with a nonlinear adaptive controller whose dynamics are a type of inverse of piezoelectric actuator. In Fig. 4, the structure of adaptive inverse control for piezoelectric actuator is shown. A particular benefit of using this scheme is that the dynamics of the closed-loop system are equal to the dynamics of the open-loop system. Therefore, instable phenomenon of the system aroused possibly by feedback scheme can be avoided.

3.2. Design of control algorithm

In this paper, we develop a nonlinear adaptive inverse controller for compensating hysteresis in piezoelectric actuator. The input of the nonlinear adaptive inverse controller is the desired displacement of piezoelectric actuator and the output is the applied voltage. The measured displacement is compared with the desired displacement, and the error is fed back to the nonlinear adaptive inverse controller through the proposed adaptive algorithm for adjusting the controller. Although hysteresis in piezoelectric actuator has no theoretical inverse, the proposed controller can still be considered as an inverse in application. Therefore, this adaptive inverse control scheme allows hysteresis in piezoelectric actuator to be canceled by its inverse. As can be seen in Fig. 4, we directly use this nonlinear adaptive inverse controller to control the piezoelectric actuator, and the dynamics modeling for hysteresis behavior in piezoelectric actuator is avoided. It means that the proposed controller can be used for various piezoelectric actuators with different dynamics. For the nonlinear adaptive inverse controller, we set the input vector as:

$$\bar{X}(k) = \{\bar{y}(k-1), \dots, \bar{y}(k-n_{\bar{y}}); \bar{y}_d(k-1), \dots, \bar{y}_d(k-n_{\bar{y}_d})\} \quad (10)$$

where $n_{\bar{y}}$ and $n_{\bar{y}_d}$ are the associate maximum lags. The ability of constructing online adaptive rate-dependent hysteresis model is the starting point of using the proposed method to construct adaptive inverse hysteresis model which can be employed as the desired controller. Similar to the proposed modeling process as shown in Section 2, it follows from (4), (5) and (6) that:

$$\bar{O}_i(k) = \bar{X}_i(k), \quad i = 1, 2, \dots, n_{\bar{y}} + n_{\bar{y}_d} \quad (11)$$

$$\bar{O}_j(k) = f \left[\sum_{i=1}^{n_{\bar{y}}+n_{\bar{y}_d}} \bar{\omega}_{ij}(k-1) \bar{O}_i(k) \right], \quad j = 1, 2, \dots, P \quad (12)$$

$$\bar{u}(k) = \sum_{j=1}^P \bar{\omega}_j(k-1) \bar{O}_j(k) \quad (13)$$

where k represents the k th sampling time; $\bar{u}$ represents the output of the proposed controller; $\bar{O}_i$ and $\bar{O}_j$ represent the output of the input layer and the hidden layer, respectively; $\bar{\omega}_{ij}$ and $\bar{\omega}_j$ represent the weights in ADALINEs in hidden layer and output layer, respectively; P represents the number of ADALINEs in the hidden layer. In

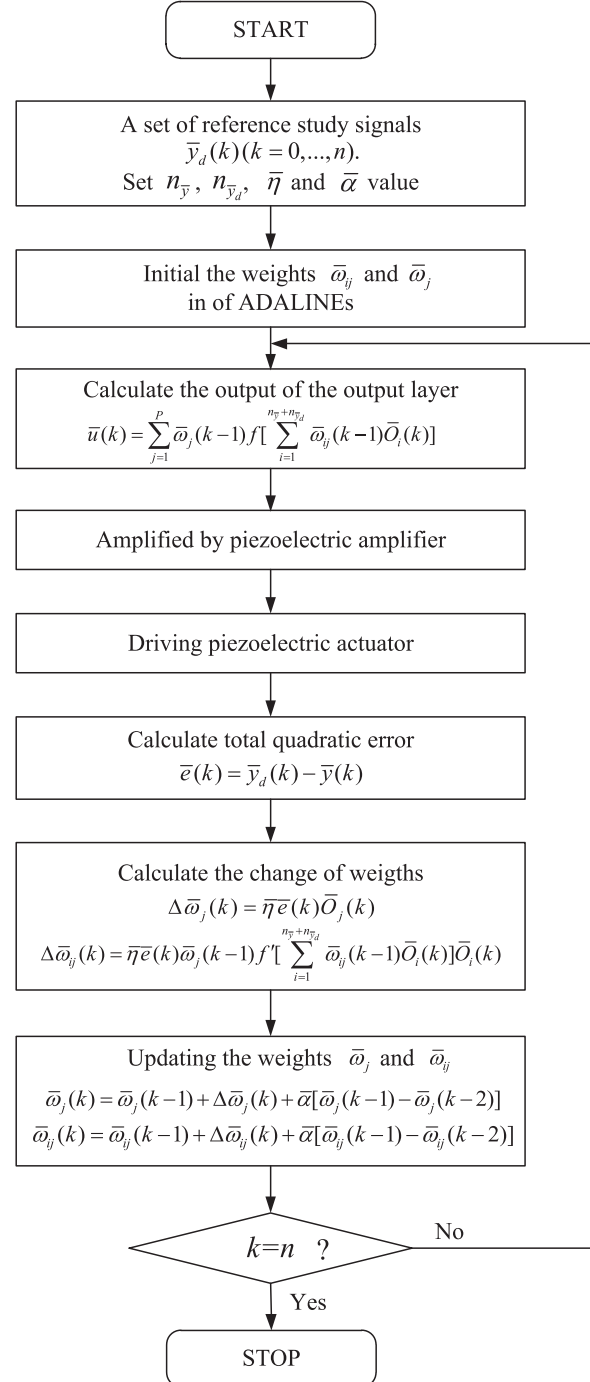


Fig. 5. The flow chart of the control algorithm.

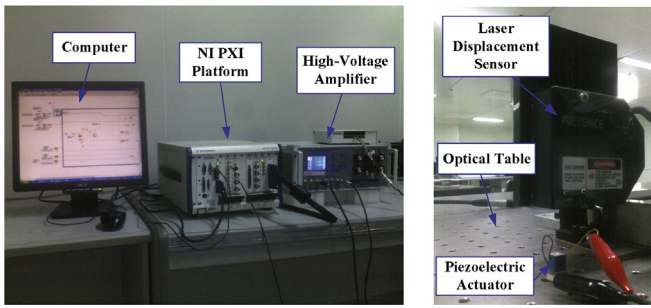


Fig. 6. The experiment platform.

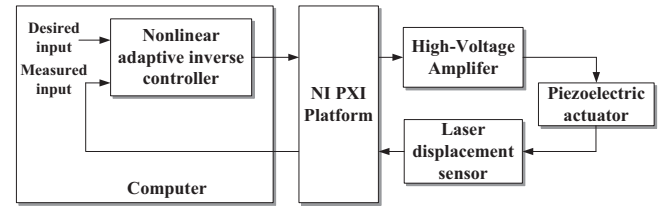


Fig. 7. The structure of the experimental system.

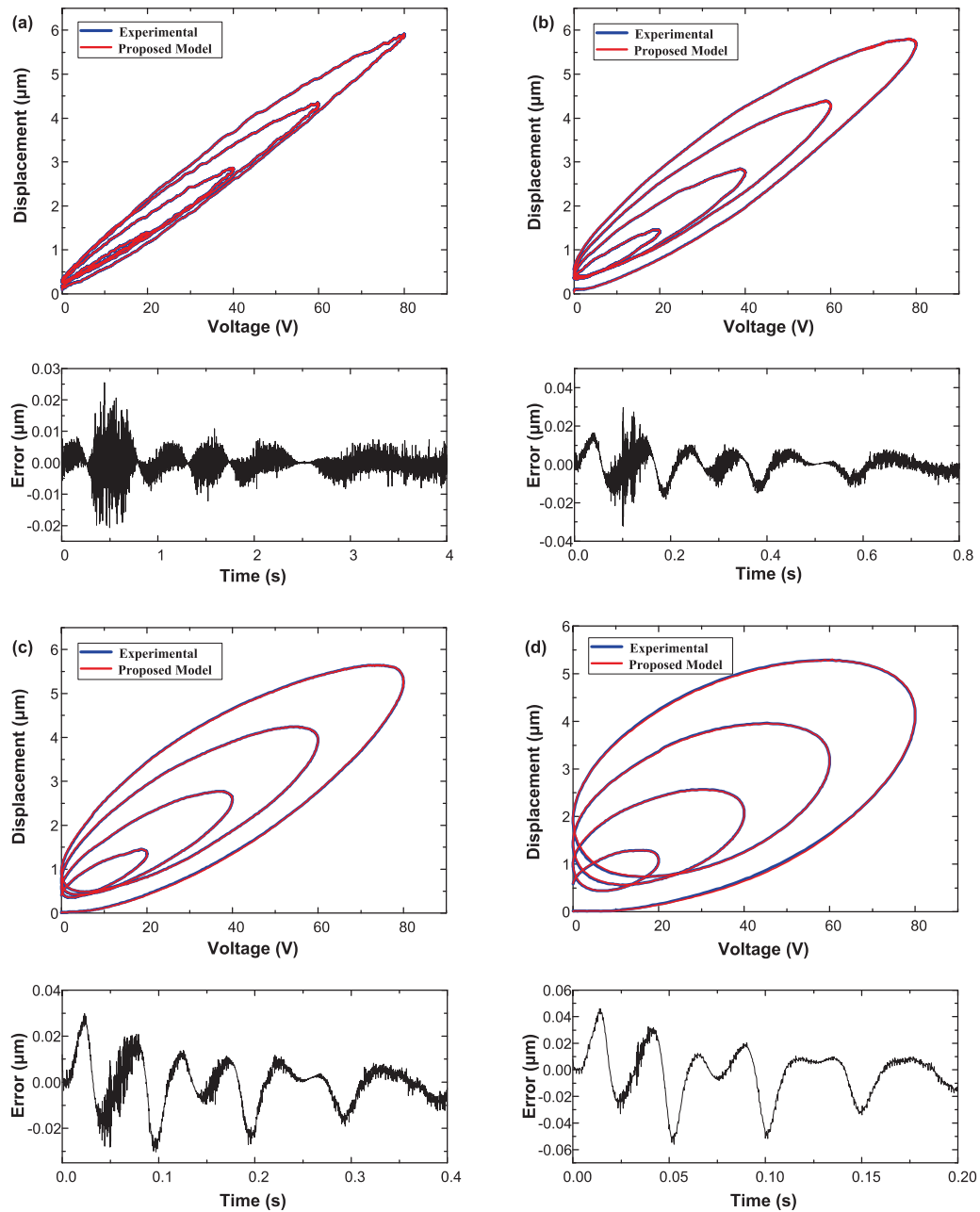


Fig. 8. Experimental verification of the proposed rate-dependent hysteresis model under excitations with different frequencies: (a) 1 Hz; (b) 5 Hz; (c) 10 Hz and (d) 20 Hz.

the practical application, $\bar{u}(k)$ needs to be amplified by a piezoelectric actuator voltage amplifier before applying to the piezoelectric actuator. After obtaining the measured displacement of piezoelectric actuator, the error between the reference displacement and measured displacement can be derived as:

$$\bar{e}(k) = \bar{y}_d(k) - \bar{y}(k) \quad (14)$$

According to (8) and (9), the weights in nonlinear adaptive inverse controller are updated as follows:

$$\bar{\omega}_j(k) = \bar{\omega}_j(k-1) + \bar{\eta} \bar{e}(k) \bar{O}_j(k) + \bar{\alpha} [\bar{\omega}_j(k-1) - \bar{\omega}_j(k-2)] \quad (15)$$

$$\begin{aligned} \bar{\omega}_{ij}(k) = & \bar{\omega}_{ij}(k-1) + \bar{\eta} \bar{e}(k) \bar{\omega}_{ij}(k-1) \cdot f' \left[\sum_{i=1}^{\bar{n}_y + \bar{n}_{y_d}} \bar{\omega}_{ij}(k-1) \bar{O}_i(k) \right] \\ & \times \bar{O}_i(k) + \bar{\alpha} [\bar{\omega}_{ij}(k-1) - \bar{\omega}_{ij}(k-2)] \end{aligned} \quad (16)$$

where $\bar{\eta}$ represents the study rate; $\bar{\alpha}$ represents the dynamic term. The flow chart of the control algorithm is shown in Fig. 5. Utilizing the strategy to adaptively updating the weights, we can adjust controller in an online way so as to realize tracking and positioning for piezoelectric actuator.

4. Experimental validation

4.1. Experimental setup

In order to validate the performance of the online adaptive rate-dependent hysteresis model and the nonlinear adaptive inverse controller, an experimental platform was established as shown in Fig. 6. The experimental platform was set up on an optical table for vibration isolation and experiments were carried out in a clean room. In this case, the piezoceramic stack actuator was glued firmly on the platform using epoxy resin. The dimensions of the stack actuator are 8 mm × 8 mm × 10 mm and its nominal displacement

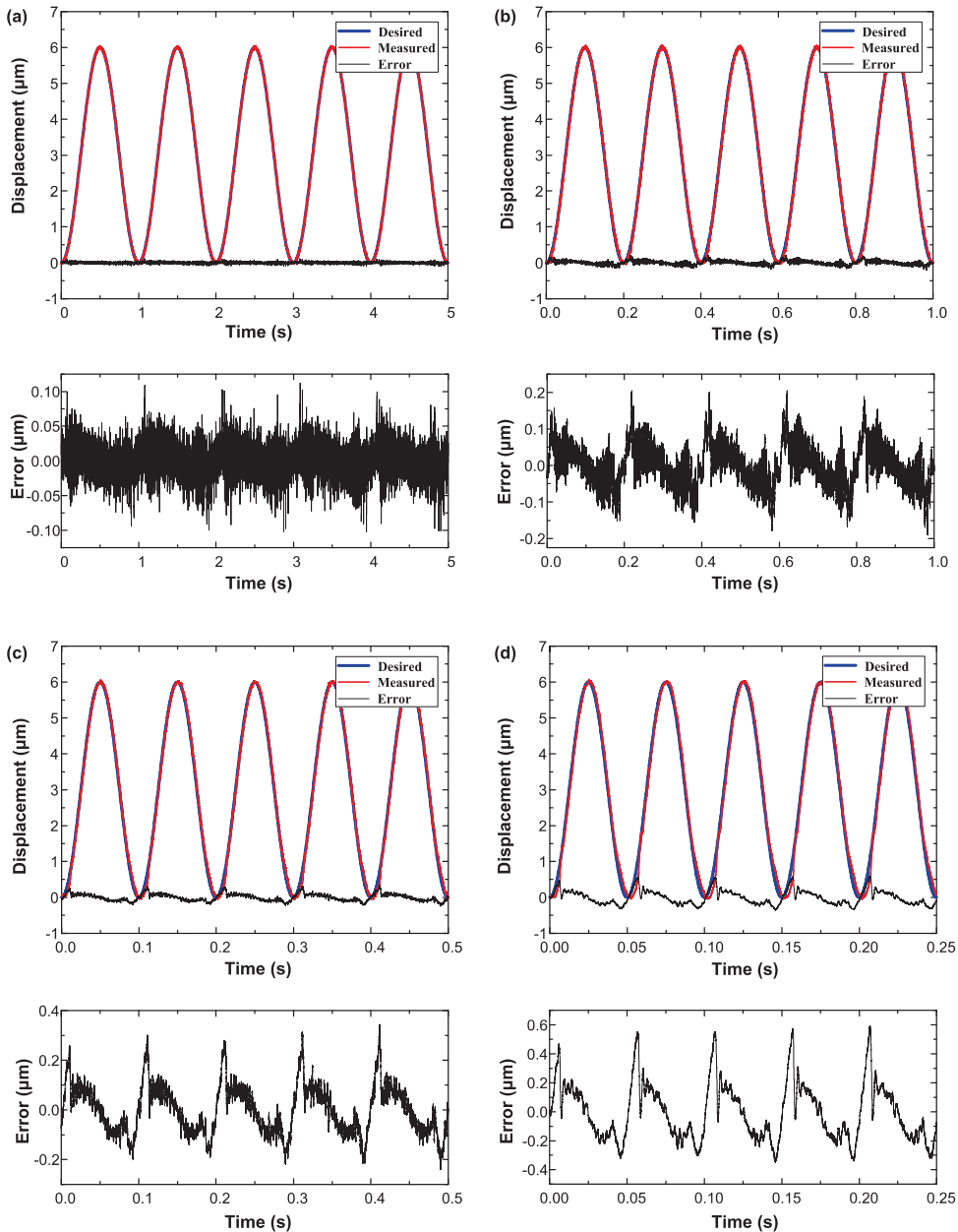


Fig. 9. Position tracking of sinusoidal waveforms with amplitude of 6 μm and frequency of: (a) 1 Hz; (b) 5 Hz; (c) 10 Hz and (d) 20 Hz.

was 20 μm for the maximum input voltage of 300 V. The actuator was vertically positioned such that the displacement of its top surface can be measured. For measuring the actual displacements of piezoelectric actuator, the CCD laser displacement sensor with 10 nm resolution (LK-G, made by Keyence Corporation) was fixed on the positioning stage (XWG40, made by Misumi Corporation). The signals were acquired by a dynamic signal acquisition module (NI PXI-4472, made by National Instruments Corporation) with 24-bit resolution ADCs and 102.4 kS/s sample rate. To achieve the real-time control, PXI Platform (NI PXI-8186, made by National Instruments Corporation) equipped with LabVIEW 8.6 system was used as the real-time control system. The LabVIEW Real-Time Module was adopted and the loop timing source was specified as 1 MHz clock. The control signals were outputted by a high-speed analog output module (NI PXI-6733, made by National Instruments Corporation) with 16-bit resolution and 10 MS/s update rate. The output signals were amplified by a high-voltage piezoelectric amplifier (Rhvd-3C, made by RongZhiNaXin Corporation) which provided excitation voltage for piezoelectric actuator from -20V to 200V with a fixed gain before applying to piezoelectric actuator. Both the input voltage and the feedback displacement were acquired and stored by a computer with Intel Core i3 2.93 GHz processor and 2G RAM. The whole structure of the experimental system is illustrated in Fig. 7.

4.2. Results and discussion

To demonstrate the performance of the online adaptive rate-dependent hysteresis model for piezoelectric actuator, hysteresis behavior in piezoelectric actuator was measured and the proposed model was constructed for comparison. In this case, we measured the first-order reversal curves which were obtained by applying sinusoidal voltage with decreasing amplitude. The major hysteresis loop of first-order reversal curves was obtained by applying sinusoidal voltage with peak-to-peak amplitude of 80 V and offset of 40 V, namely $U = 40 + 40 \sin(2\pi ft - \pi/2)$, where f represents the exciting frequency. The minor loops were obtained by decreasing the amplitude of applied voltage by 20 V for each hysteresis loop. Experimental verification of the proposed rate-dependent hysteresis model under excitations with different frequencies are shown in Fig. 8. It can be seen that the shapes of hysteresis loops change with both the amplitude and frequency of applied voltage. On the one hand, hysteresis decreases with the decreasing amplitude of applied voltage; on the other hand, hysteresis increases with the increasing frequency of applied voltage. Although hysteresis in piezoelectric actuator is both amplitude-dependent and rate-dependent, it is shown that the proposed adaptive model is capable of describing hysteresis behavior in piezoelectric actuator accurately.

For the purpose of validating the performance of the nonlinear adaptive inverse controller, we employed this controller to control piezoelectric actuator for tracking sinusoid signals with frequencies of 1 Hz, 5 Hz, 10 Hz and 20 Hz, as shown in Fig. 9. The input signals were sinusoidal waveforms with peak-to-peak amplitude of 6 μm and offset of 3 μm , that is $y_d = 3 + 3 \sin(2\pi f_d t - \pi/2)$, where f_d is the frequency of input signal. Although the piezoelectric actuator has rate-dependent hysteresis characteristic which result in evident discrepancy in hysteresis loops with different input frequencies as shown in Fig. 8, the proposed controller renders a much better position tracking performance for 1 Hz and 5 Hz and moderately better performance for 10 Hz and 20 Hz as well. The compensated hysteresis curves with different frequencies are shown in Fig. 10. Due to the compensation brought by the nonlinear adaptive inverse controller, hysteresis behavior in piezoelectric actuator is strongly reduced.

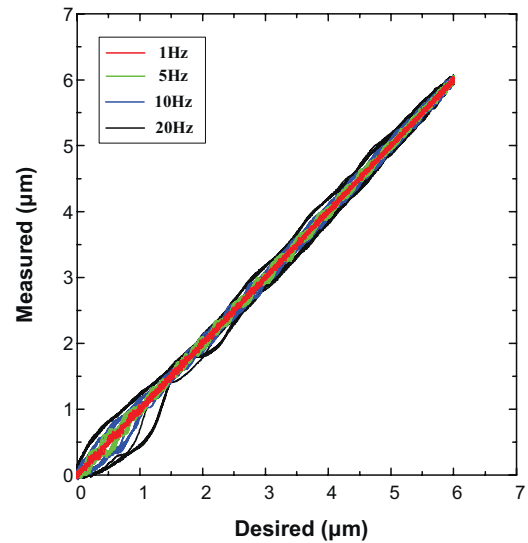


Fig. 10. The compensated hysteresis curves with different frequencies.

Piezoelectric actuator is widely used at low frequency, especially in ultra-precision positioning. For instance, most atomic force microscopes (AFMs) are driven by piezoelectric actuators. A critical cause for loss in precision is drift due to the creep effects of piezoelectric actuator. Thus, we conducted an experiment to demonstrate the static positioning performance of the nonlinear adaptive inverse controller. The input signal was square waveform with changed amplitude. The amplitude increased from 0 μm to 6 μm and then decreased from 6 μm to 0 μm . As shown in Fig. 11, taking advantage of the proposed controller, creep effect in piezoelectric actuator can also be restrained.

Furthermore, in many industrial applications, the desired motion signals of piezoelectric actuator are non-periodic. In this case, the proposed nonlinear adaptive inverse controller was tested with randomly created desired motion to demonstrate its tracking

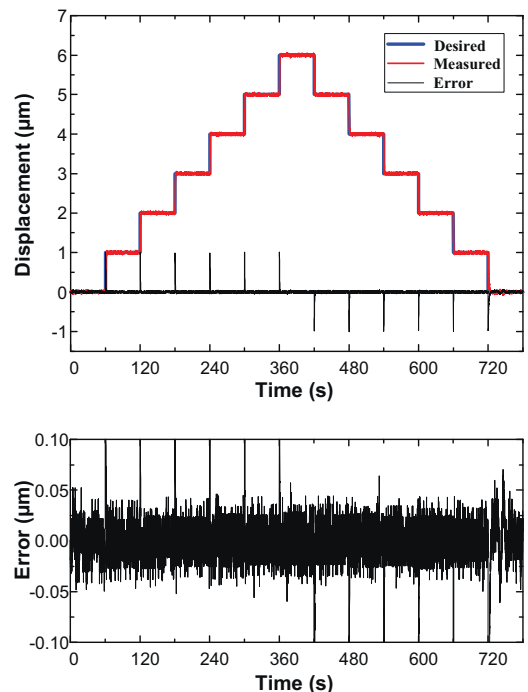


Fig. 11. Static position tracking of square wave with changed amplitude.

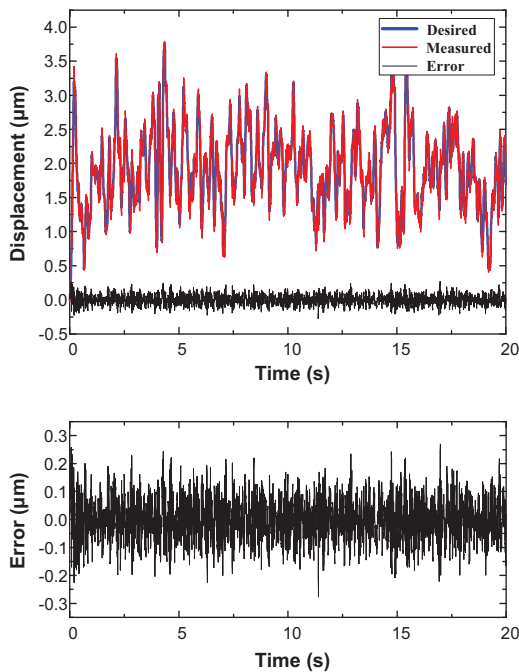


Fig. 12. Position tracking of randomly created non-periodic wave.

Table 1
The RMSE of tracking and positioning experiments.

Waveform	RMSE (μm)
1 Hz sinusoidal wave	0.0248
5 Hz sinusoidal wave	0.0579
10 Hz sinusoidal wave	0.0996
20 Hz sinusoidal wave	0.1987
Square wave	0.0218
Random wave	0.0722

ability for non-periodic input signal. This desired motion was created by uniform white noise passed through a Butterworth filter using LabVIEW 8.6 system. As shown in Fig. 12, the experimental result demonstrates that the proposed controller is competent for tracking random input motions signal. In Table 1, the RMSE of all these experiments are listed for comparison.

5. Conclusion

In this paper, we propose an online adaptive rate-dependent hysteresis model to accurately describe hysteresis behavior in piezoelectric actuator. In comparison with hysteresis model constructed and identified in an offline way, the proposed online model features of both the high precision and high efficiency, allowing it to be implemented in real-time control system. Furthermore, the proposed model exhibits excellent rate-dependent property which agrees well with real hysteresis behavior in piezoelectric actuator. Aiming at effective compensation of hysteresis in piezoelectric actuator, we proposed a nonlinear adaptive inverse controller. One distinctive feature of the proposed controller is that the controller can be designed even the dynamics model of piezoelectric actuator is unknown. It means that the proposed controller can be considered as a general controller for various piezoelectric actuators with different dynamics. The actual implementation of the proposed controller was carried out on a piezoceramic stack actuator. Experimental results show that the hysteresis in piezoelectric actuator were obviously restrained, allowing the positioning and tracking performance to reach as precision as ten nanometer order. This can be considered as the accuracy limitation subjected to

the resolution of measurement system and environmental noise. The rate-dependent property of the proposed controller is verified by tracking input signals with different frequencies. Furthermore, another advantage of the proposed controller is that it is capable of tracking non-periodic random motion. More importantly, because the control system is open-loop in essence, the unstable phenomenon aroused by feedback can be avoided. Through a series of experimental examinations, it is proved that the proposed controller can be effectively applied to the motion stage using piezoelectric actuator to achieve ultra-precision motion control.

Acknowledgments

The research is gratefully supported by the National Natural Science Foundation of China (Grant No. 51235005), the Foundation for Innovative Research Groups of the National Natural Science Foundation of China (Grant No. 51121002), the Major State Basic Research Development Program of China (973 Program) (Grant No. 2009CB724205).

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